When Search Is Not Enough: Diagnosing Reasoning Failures and Constructing Verifiable Executable Capabilities

Chimdi Walter Ndubuisi University of Missouri

CORA on ARC-AGI-2 · development results · ARC Prize Research Summit, 23 October 2026

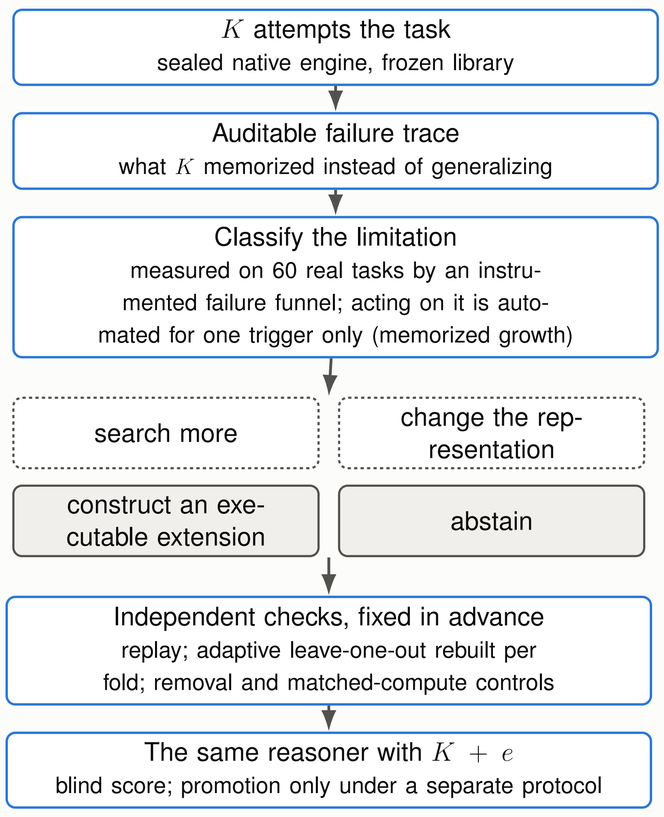
The question

How can a reasoning system know when its existing way of solving problems is inadequate, and acquire a better one without compromising the evidence used to judge it?

A failed solve can need different responses:

- **search correction**: the answer is expressible, search missed it
- **representation correction**: the needed view was screened out
- **executable capability construction**: the language lacks it
- **evidence insufficiency**: the examples underdetermine the rule
- **certification failure**: a valid candidate fails the checks

The CORA loop



Solid: implemented and measured. Dashed: proposed, not automated. A general failure-type controller is **proposed, not implemented**.

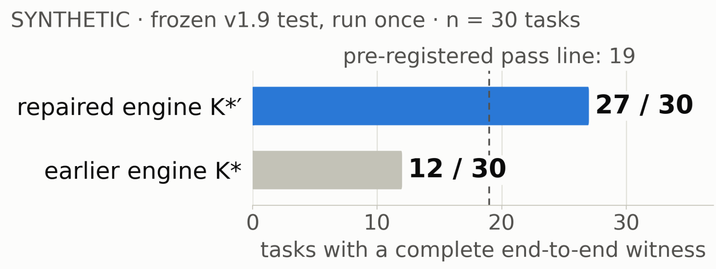
Four claims, four kinds of evidence

(a) the extension causes the solve	development only: nothing accepted without it, or with matched compute; not certified
(b) the real failure trace matters	demonstrations-only and shuffled failures never gave an accepted, used extension (4 tasks)
(c) a bounded semantic extension	**not established**
(d) retention and later reuse	**untested**

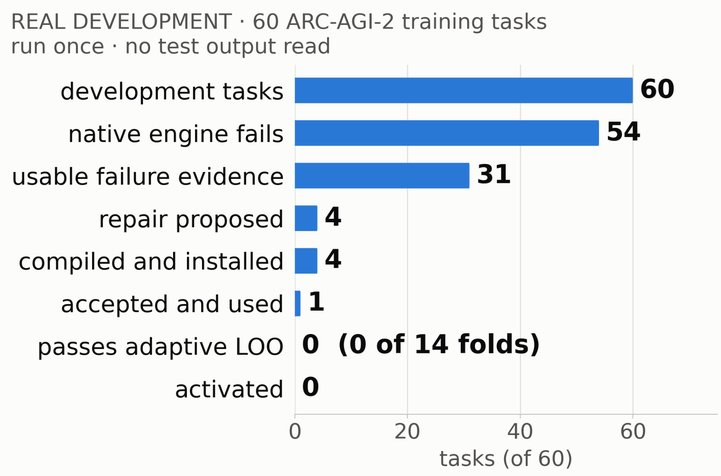
Repairs tested before use

- Forcing a view to be searched first: no task gained, 5 of 24 synthetic tasks harmed (4 correct answers lost, 1 false acceptance). **Not run.**
- Fold-local removal of duplicate views: gained 1, harmed 4 of 41. **Rejected.**

What has been measured

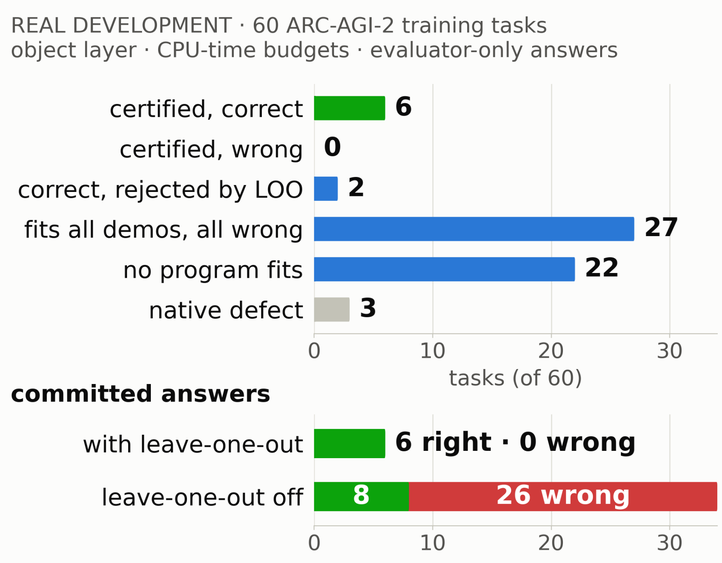


A synthetic claim only. The repaired engine also accepted every deliberately wrong extension it was given (5 of 5, 28 of 28).



One constructed extension was accepted and used by the unchanged engine; none survived adaptive leave-one-out. **No new ARC solve**: the certified training score stays 185/1,000.

Where CORA fails on real tasks



The checks are not the problem: nothing wrong was certified, and switching leave-one-out off would add 2 right answers and 26 wrong ones. **The rules are**: in the 27 "fits all demos" tasks (26 with CORA's frozen library loaded) the programs memorize. 21 of 25 use constant parameters, mostly a grown cell pattern keyed to a colour or an exact shape, and each leave-one-out fold rebuilds a different one (59 of 74 failed folds).

- Native's own relational re-search finds no program (0 of 11).
- Three times the search budget fixes none (0 of 14).
- Given the true answer as a demonstration (diagnostic only), it fits 26 of 27, in the same view in 24, by memorizing one more case.
- The view-admission failure we first traced explains only 7 of 76 failed folds.

Frozen neural guides did not pick views better than the fixed order (Qwen2.5-1.5B 19 of 35, I-JEPA 3 of 35; pre-registered, synthetic).

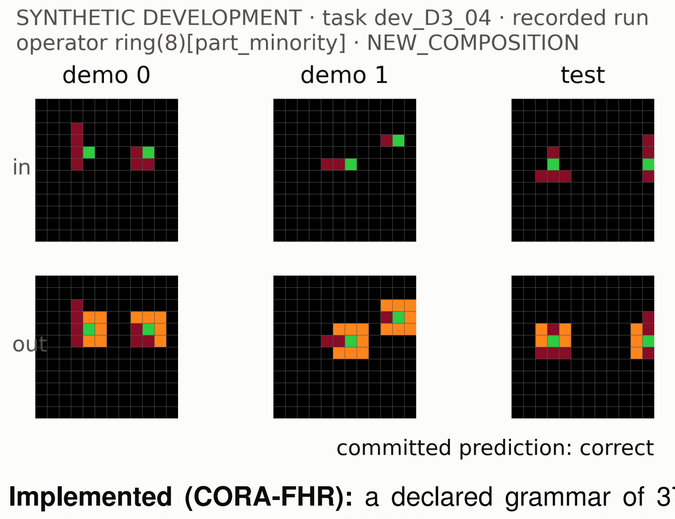
Competition status, 14 October

OFFICIAL KAGGLE · native CORA v23: **0.00**.

NEURAL_TTT_BASELINE · a reproduction of the public NVARC Qwen3-4B test-time-training pipeline, **not CORA**: official public score **27.64**, equal to NVARC's own 2025 public score. On the 240-task stand-in it decoded every task (259 of 259 genuine predictions, 0 fallbacks) in 4.3 h.

DEVELOPMENT · whole native harness: 12 of 60; 18 of its 29 claimed answers were wrong, because the pipeline layer answers first. Reordering the answers it already produces (shadow study): 14 of 60, and 36 to 39 of 60 on synthetic C2-dev, none lost. Engineering only; not submitted.

Constructing the missing operation SYNTHETIC DEVELOPMENT

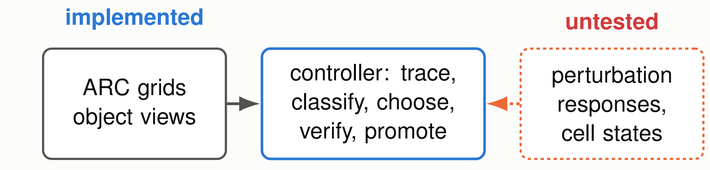


committed prediction: correct

Implemented (CORA-FHR): a declared grammar of 37,750 typed growth operators; CORA's own memorized-growth record as evidence; reversible installation into the unchanged engine; leave-one-out that rebuilds the operator in every fold.

- **One complete witness.** Frozen CORA fails. The constructed one-step operator (a ring around the minority-colour cell) is new: on 300 perturbed shapes native reproduces it 7 times, the best existing constructor 8. The task's planted second step adds nothing on its shapes, so this is not a two-step rule. All 3 folds rebuild it and predict exactly. The committed answer is correct, and removing the operator removes the gain.
- 27 development tasks: 0 wrong certificates, 0 lost answers. Matched extra search: 0 of 15.
- **Not yet shown:** that the failure record is what matters. A failure-blind constructor found the same operator, and on this corpus its evidence equals the authentic arm's by design. The prospective test is sealed.

A portable controller? BIOLOGICAL APPLICATION: UNTESTED



Portable: the repair-and-verification contract, not the ARC operators. A test needs held-out perturbations and cell states, and controls. No biological data or result exists here.

Not claimed · evidence

No CORA solve or score gain on ARC-AGI-2; no semantic invention; no automatic failure-type controller; no evidence yet that failure records cause a gain; the neural score is not a CORA result; no domain generality; no results from the separate VDCG project. The 120-task pilot is frozen and **has not been run**.

Every number, with commit and hash: `records/CORA_SUMMIT_POSTER_EVIDENCE_20261012.md` and `records/CORA_FAILURE_FUNNEL_20261014.md`; figure hashes: `figures/PROVENANCE.json`.